

NLP Sentiment Analysis

1. Business Understanding

Objective: Understand public perception of Apple and Google by classifying tweets as negative, neutral, or positive.

Why it matters: Enables marketing and product teams to monitor brand reputation, spot issues quickly, and measure campaign impact.

2. Data Overview

- **Source:** Labeled dataset of tweets mentioning Apple/Google with three sentiment labels.
- **Distribution:** Majority neutral and positive; negative tweets are rare (class imbalance).
- **Cleaning:** Removed links, mentions, hashtags, punctuation; lowercased text and filtered common words.

3. Data Preparation

- Deep cleaning: Removed noise (links, mentions, hashtags, special characters).
- Lowercased and removed stopwords (NLTK).
- Filtered to keep only alphabetic content.
- TF-IDF vectorization (scikit-learn, 5,000 features) to capture term importance and reduce dimensionality.

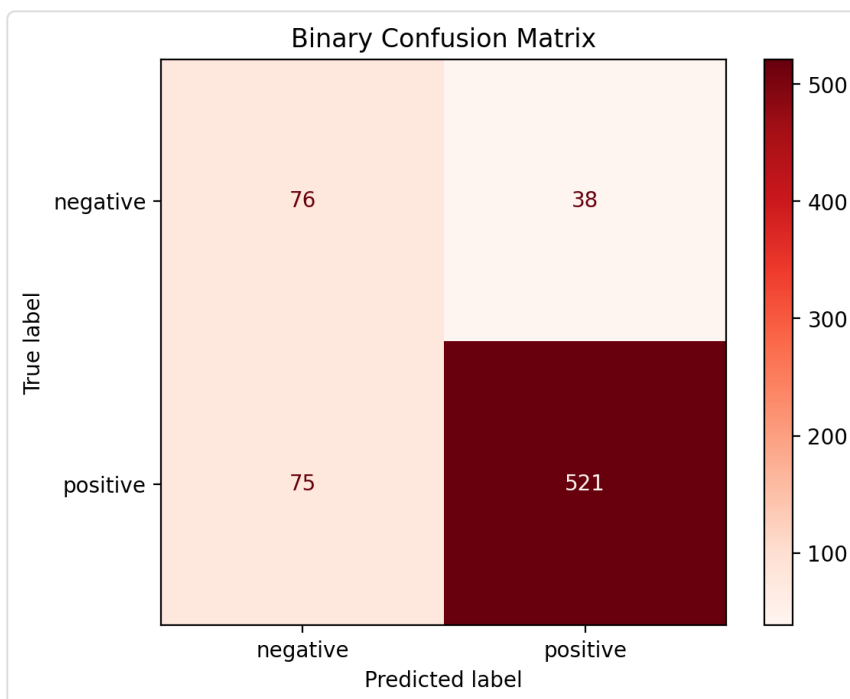
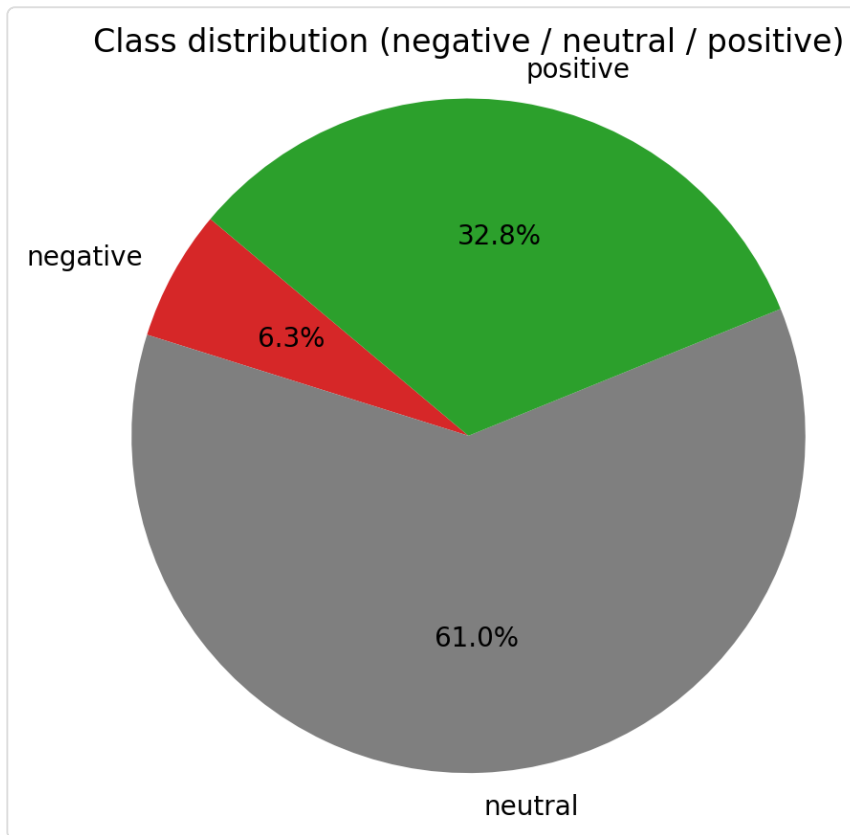
4. Modeling Approach

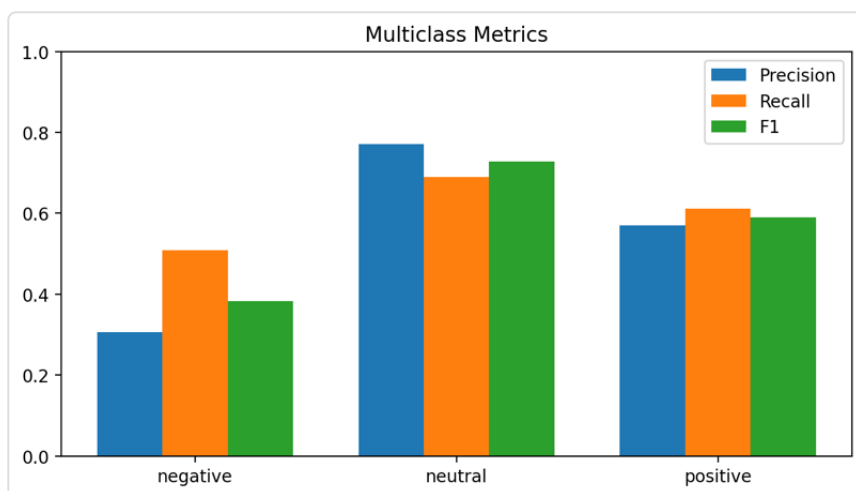
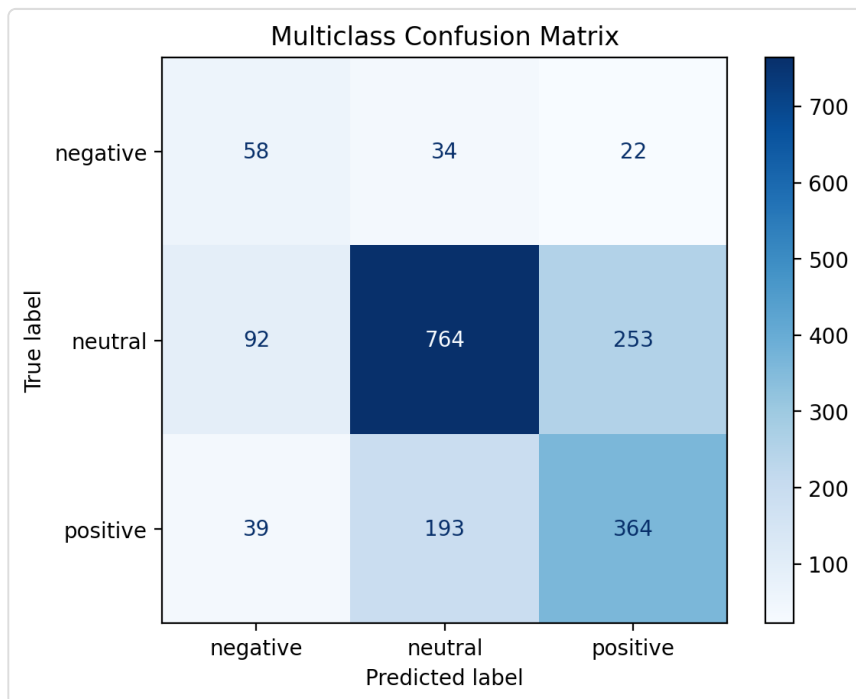
- Converted text to numeric representation (TF-IDF).
- Modeled with Logistic Regression (robust, interpretable).
- Two views:
 - Binary model (positive vs negative)
 - Multiclass model (negative, neutral, positive)
- Stratified train/test split to preserve class distribution.

5. Evaluation & Results

- Metrics: accuracy, precision, recall, F1-score, confusion matrices.
- **Key takeaways:**

- Positive and neutral tweets are easier to predict; negative is harder (few examples, ambiguous language).
- Errors mostly between neutral/positive and negative/neutral.
- Business can tune the model to maximize recall for negatives (catch more alerts).





How to Read the Metrics

- **Accuracy:** % of tweets classified correctly.
- **Precision:** Of tweets predicted as "negative", how many truly are negative?
- **Recall:** Of all true negatives, how many did the model catch?
- **F1-score:** Balance of precision and recall, useful for imbalanced classes.

Confusion Matrix Explained

- Table comparing predictions vs actual labels.
- More values on the diagonal = better performance.
- Off-diagonal cells show specific confusions (e.g., negative → neutral).

6. Feature Insights

- Each word gets a weight per class (Logistic Regression).
- Positive weight: pushes tweet toward the class; negative weight: pulls away.
- Top words per class validate model logic and spot biases.

7. Business Applications

- **Monitoring:** Alerts for spikes in negative sentiment, tracking by product/campaign/geography.
- **Customer feedback:** Route negative tweets to support/PR.
- **Campaign measurement:** Compare sentiment before/after launches, monitor competitors.

8. Limitations

- Imbalanced data: fewer negatives, harder detection.
- Nuance/context: sarcasm, irony, evolving slang.
- Domain drift: vocabulary/topics change, models need updates.

9. Recommendations & Improvements

- **Data:** Collect more negatives, label recent tweets.
- **Features:** Add bigrams, n-grams, emojis, lemmatization.
- **Models:** Try SVM, Naïve Bayes, lightweight transformer (DistilBERT).
- **Thresholds:** Tune decision thresholds for business priorities.

10. Stakeholder Summary

- Automatic tweet classification to track brand perception.
- Model performs well for positive/neutral, fast and scalable.
- Targeted improvements (more negatives, advanced features, threshold tuning) will better capture actionable signals.

11. Appendix: Technical Glossary

- **TF-IDF:** Text representation by word rarity.

- **Logistic Regression:** Interpretable, widely used classifier.
- **Precision/Recall/F1:** Performance metrics per class.
- **Confusion Matrix:** Table of confusions to guide improvements.